

Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
1	(a)	(i)		9		1		1		
		(ii)		The chromosomes have not paired up / Not (arranged) in homologous pairs/ Ignore homologous pairs unqualified Absence of bivalents/ Synapsis does not occur Ignore reference to chiasmata			1	1		
		(iii)		stains the {DNA/ nucleic acids/ histones}	1			1		1
	(b)	(i)		Anther/ ovary	1			1		1
		(ii)		C BGEH AFD		1		1		
		(iii)		Any two (x1) from - In E {chromosomes/ homologous pairs} are {segregating / being pulled to opposite poles} whereas in F the chromatids are being pulled to opposite poles (1) - In E the centromeres do not divide, in F they do divide (1) - Correct reference to the spindle (1) - E is diploid and F is haploid (1) Ignore reference to the production of cells at the end of meiosis I and meiosis II		2		2		
		(iv)		{Chiasmata / chiasma formation / crossing over/ or description of} is taking place in {photomicrograph B/ prophase 1} (1) Ignore reference to photomicrograph G {Independent assortment/random assortment/ or description of} is occurring in {photomicrograph G/ metaphase 1} (1) - Genetic variation allows {the species to adapt / some of the offspring to survive} (in changeable environments.) (1) Reject reference to {them/ offspring} {adapting}	1	2		3		
				Question 1 total	3	6	1	10	0	2